

YUXUAN YANG

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Education

Southwestern University of Finance and Economics (SWUFE)

Bachelors of Engineering in Computer Science

Chengdu, Sichuan, China

August 2018-June 2022

Washington University in Saint Louis (WUSTL)

Master of Science in Computer Science

Saint Louis, MO

August 2023-June 2025

Washington University in Saint Louis (WUSTL)

PhD in Computer Science

Saint Louis, MO

August 2025-Present

Work Experience

Southwestern University of Finance and Economics

Research Assistant for SWUFE MLDE Group

Chengdu, Sichuan, China

August 2022-June 2023

- Conducted research under professors' guidance.
- Developed a code hub for continual learning baselines with over 2000 lines of code.
- Trained over ten first-year master and undergraduate students in fundamental deep learning.

Research Projects

Learning Control Barrier Functions from Expert Demonstrations using Inverse Constraint Learning

Researcher with Prof. Hussein Sibai

January 2025-September 2025

- Proposed a data-driven framework that learns neural Control Barrier Functions (CBFs) from expert demonstrations when the failure set is hard to specify formally. Used Inverse Constraint Learning (ICL) to infer a constraint classifier from demonstrations, then leveraged it to label simulated trajectories for neural CBF training. Evaluated on four environments including autonomous driving scenarios, outperforming existing baselines and matching ground-truth-labeled CBF performance.

Designing Latent Safety Filters using Pre-Trained Vision Models

Researcher with Prof. Hussein Sibai

April 2025-September 2025

- Investigated the use of pre-trained vision models (PVRs) as perception backbones for vision-based safety filters. Applied PVRs to failure set classifiers and Hamilton-Jacobi reachability-based safety filters with latent world models. Analyzed trade-offs between training from scratch, fine-tuning, and freezing PVR weights across robotics control settings.

Forgetting Prevention for Cross-regional Fraud Detection with Heterogeneous Trade Graph

Researcher with Prof. Xin Yang

December 2021-March 2022

- Proposed a novel solution based on heterogeneous trade graphs, namely Heterogeneous Trade Graph learning for Cross-regional Fraud Detection (HTG-CFD), to prevent knowledge forgetting of cross-regional fraud detection and continuously detect fraudulent transactions in the process of business expansion in new regions.

Publications

- **Yuxuan Yang**, Hussein Sibai, "Learning Neural Control Barrier Functions from Expert Demonstrations using Inverse Constraint Learning," *ICRA*, 2026.
- Ihab Tabbara*, **Yuxuan Yang***, Ahmad Hamzeh, Maxwell Astafyev, Hussein Sibai, "Designing Latent Safety Filters using Pre-Trained Vision Models," *ICRA*, 2026. (*equal contribution)
- Xin Yang, Yujie Li, Dan Meng, **Yuxuan Yang**, Dun Liu, Tianrui Li, "Three-way Multi-granularity Learning towards Open Topic Classification," *Information Sciences*, 2022.
- Xin Yang, **Yuxuan Yang**, Junfang Luo, Dun Liu, Tianrui Li, "A Unified Incremental Updating Framework of Attribute Reduction for Two-dimensionally Time-evolving Data," *Information Sciences*, 2022.